

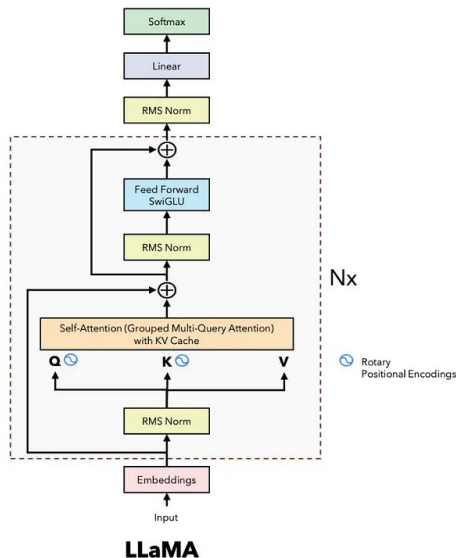
Delta Quantization for DLMs

Ahmad Fraij

Llada Architecture and Submodules

- There are 7 linear projections per block but only 4 unique inputs:
 - qkv_in: q_proj, k_proj and v_proj share same input (attn_norm output)
 - attn_out_in: attn_out receives result of attention after concatenating heads
 - ff_in: ff_proj and up_proj both share same input (ff_norm output)
 - ff_out_in: $\text{SiLU}(\text{ff_proj}) \odot \text{up_proj}$

```
LLaDAModelLLM(  
  (model): LLaDAModel(  
    (transformer): ModuleDict(  
      (wte): Embedding(126464, 4096)  
      (emb_drop): Dropout(p=0.0, inplace=False)  
      (ln_f): RMSLayerNorm()  
      (blocks): ModuleList(  
        (0-31): 32 x LLaDALlamaBlock(  
          (dropout): Dropout(p=0.0, inplace=False)  
          (act): SiLU()  
          (attn_out): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (ff_out): SINQLinear(in_features=12288, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (rotary_emb): RotaryEmbedding()  
          (attn_norm): RMSLayerNorm()  
          (ff_norm): RMSLayerNorm()  
          (q_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (k_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (v_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (ff_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cpu, W_q.device=cpu)  
          (up_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cpu, W_q.device=cpu)  
        )  
      )  
      (ff_out): Linear(in_features=4096, out_features=126464, bias=False)  
    )  
  )  
)
```



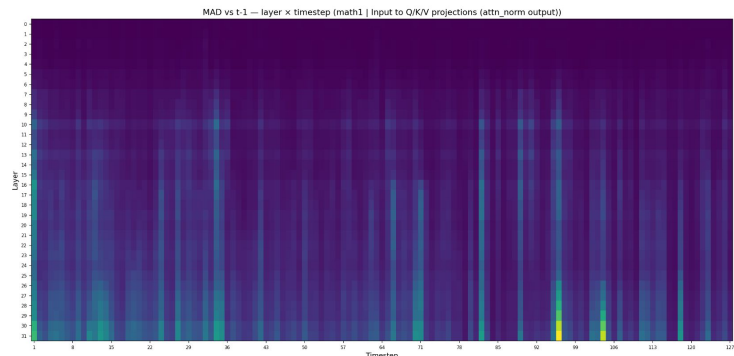
Metric for Analysis

Activation analysis metrics. $\mathbf{a}_t \in R^d$ denotes the mean activation vector at timestep t (averaged over batch and sequence dimensions), $\delta_t = \mathbf{a}_t - \mathbf{a}_{t-1}$ denotes the activation delta, and $\kappa(\mathbf{v}) = \frac{E[(v-\mu)^4]}{\sigma^4} - 3$ denotes excess kurtosis computed on the flattened tensor.

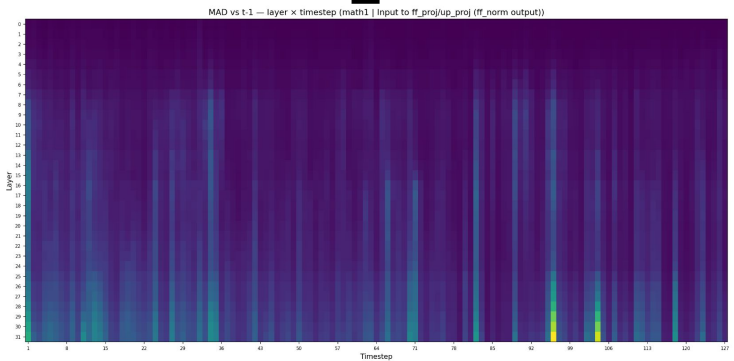
Metric	Equation
Mean Absolute Difference (MAD)	$\frac{1}{d} \sum_{i=1}^d \delta_{t,i} $
Relative Mean Absolute Difference (Rel. MAD)	$\frac{\text{MAD}_t}{\frac{1}{d} \sum_i a_{t-1,i} }$
Maximum Absolute Difference (Max Diff)	$\max_i \delta_{t,i} $
Relative Maximum Absolute Difference (Rel. Max Diff)	$\frac{\max_i \delta_{t,i} }{\max_i a_{t-1,i} }$
Activation Kurtosis	$\kappa(\mathbf{a}_t)$
Delta Kurtosis	$\kappa(\delta_t)$
Kurtosis Ratio	$\frac{\kappa(\delta_t)}{ \kappa(\mathbf{a}_{t-1}) }$

Mean Absolute Difference (MAD)

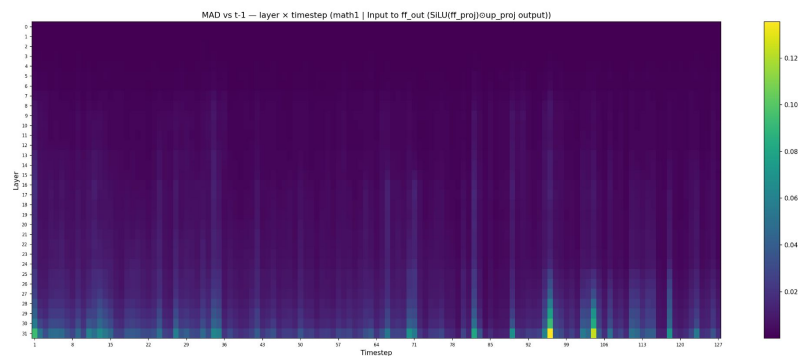
qkv_in



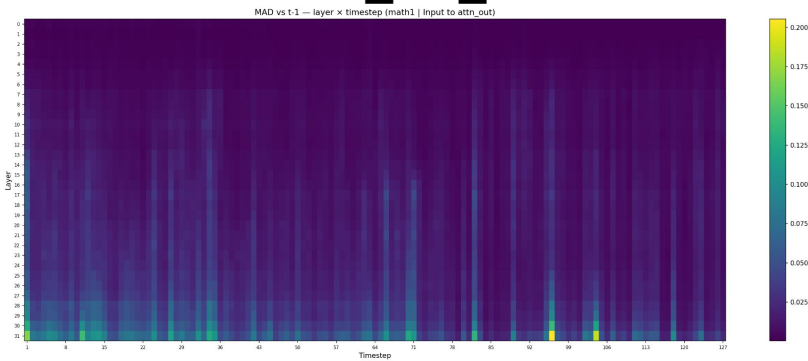
ff_in



ff_out_in

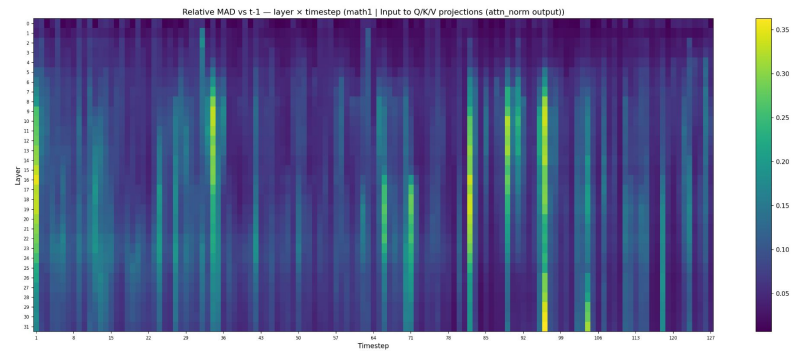


attn_out_in

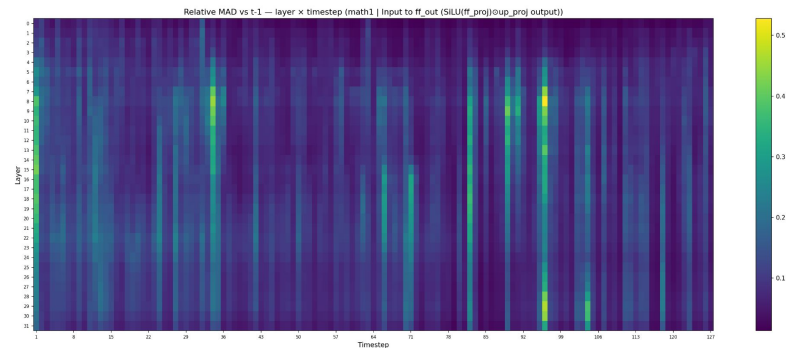


Relative Mean Absolute Difference

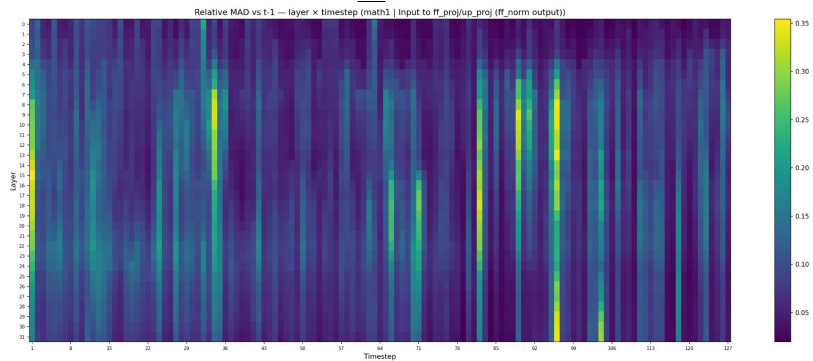
qkv_in



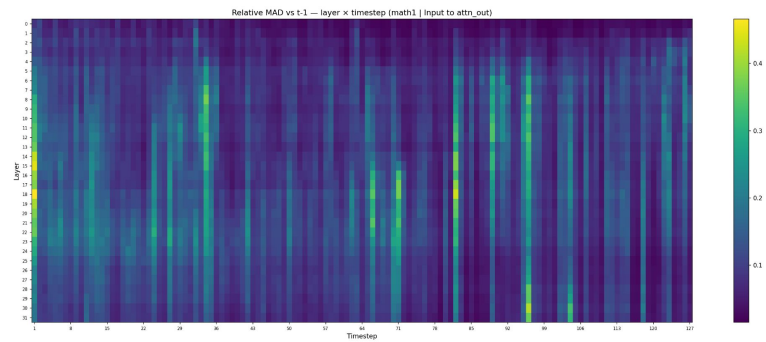
ff_out_in



ff_in

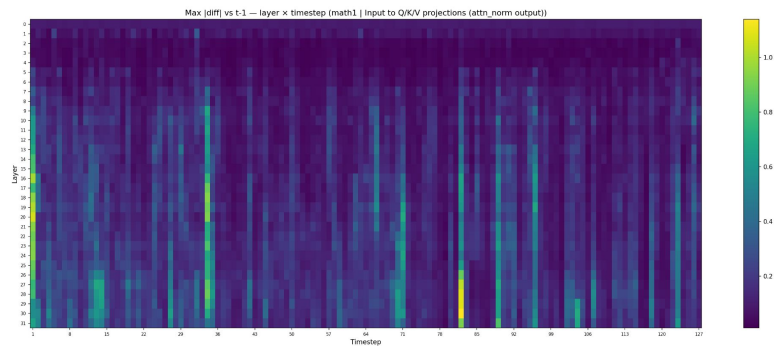


attn_out_in

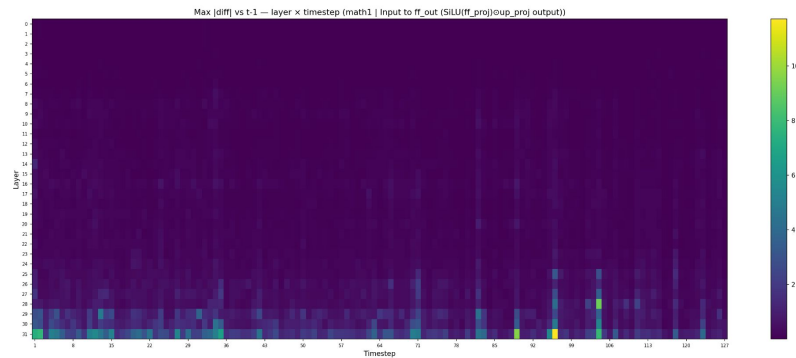


Max Absolute Difference

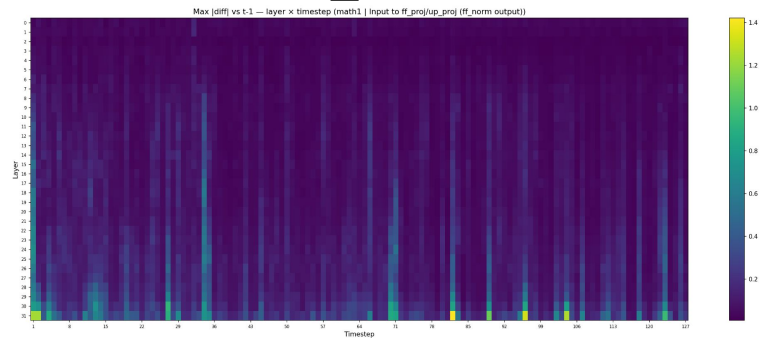
qkv_in



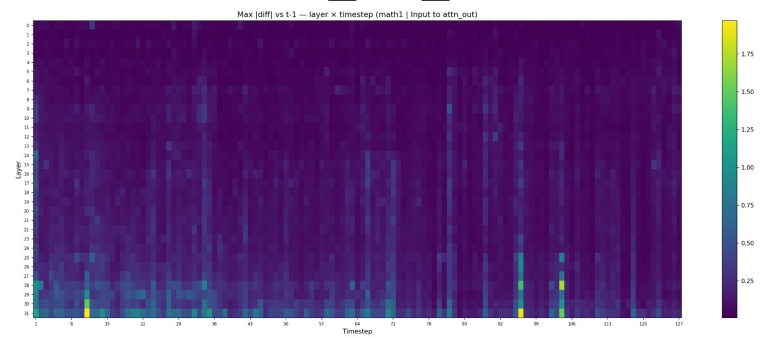
ff_out_in



ff_in

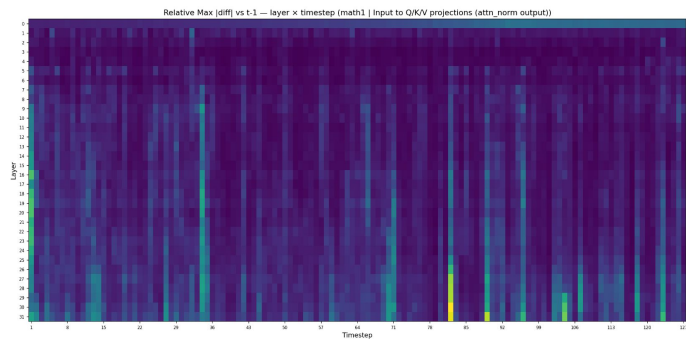


attn_out_in

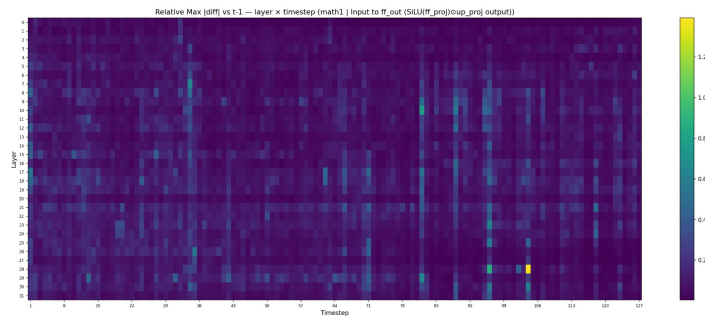


Relative Max Absolute Difference

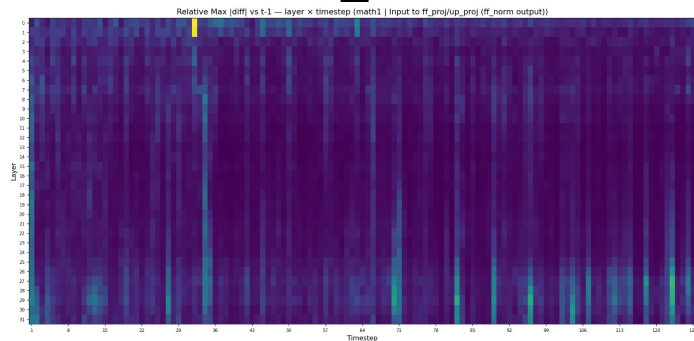
qkv_in



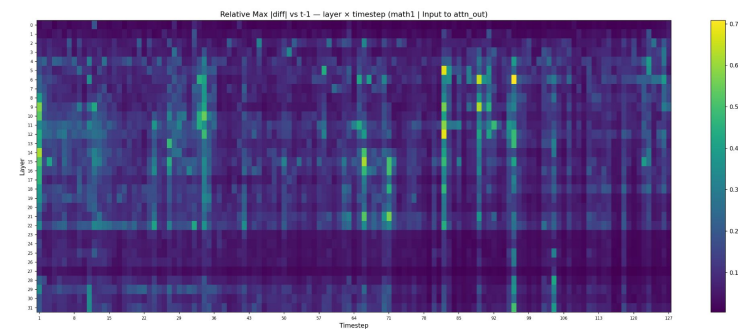
ff_out_in



ff_in

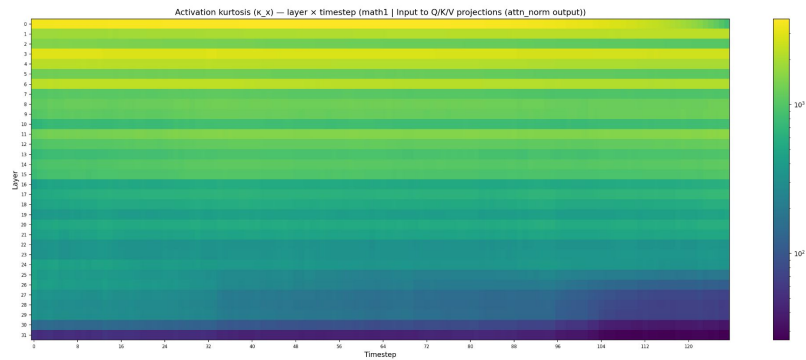


attn_out_in

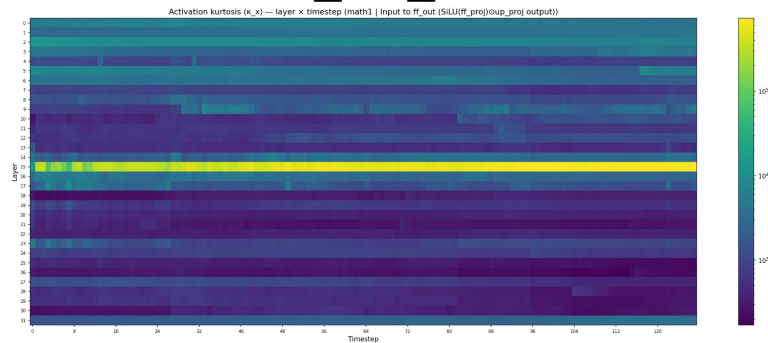


Activation Kurtosis

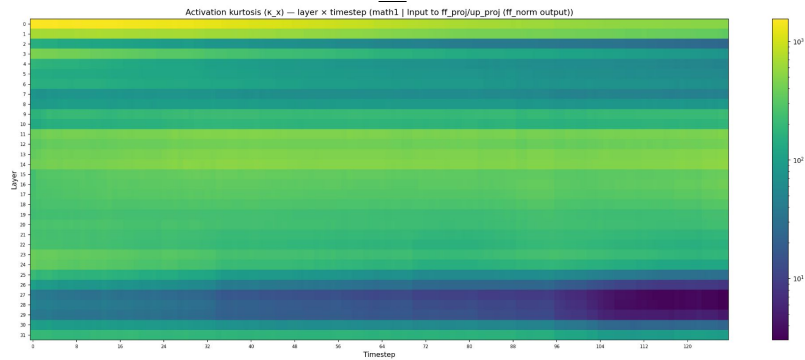
qkv_in



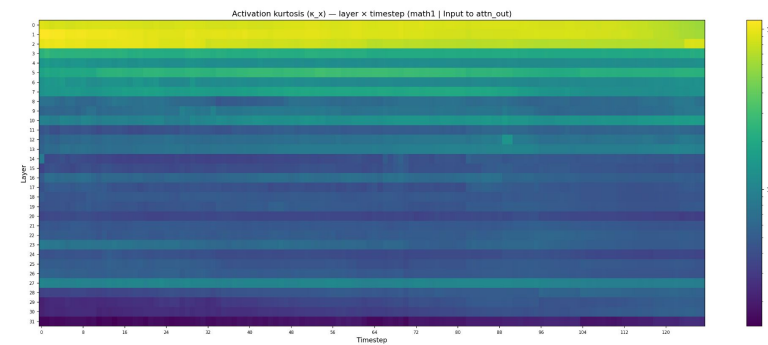
ff_out_in



ff_in

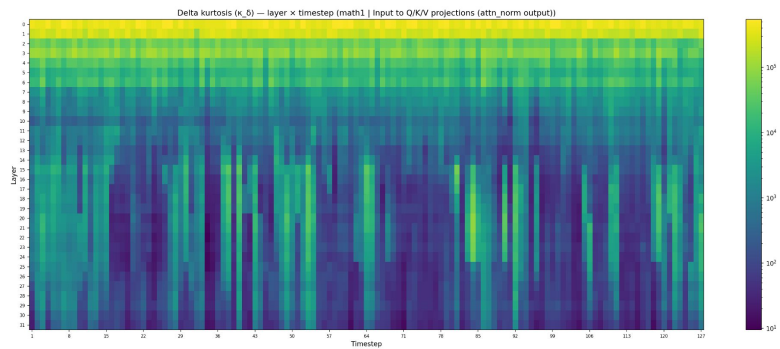


attn_out_in

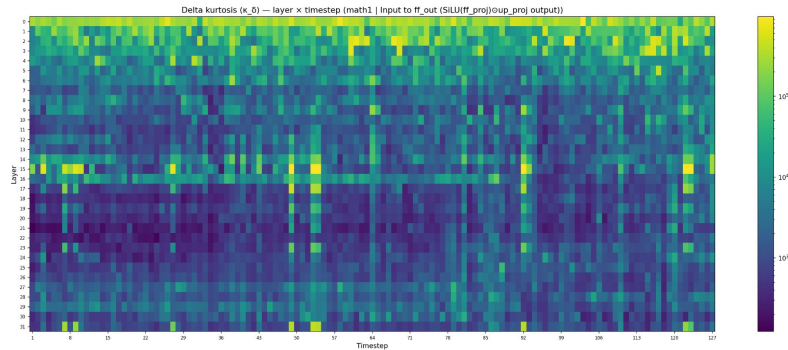


Delta Kurtosis

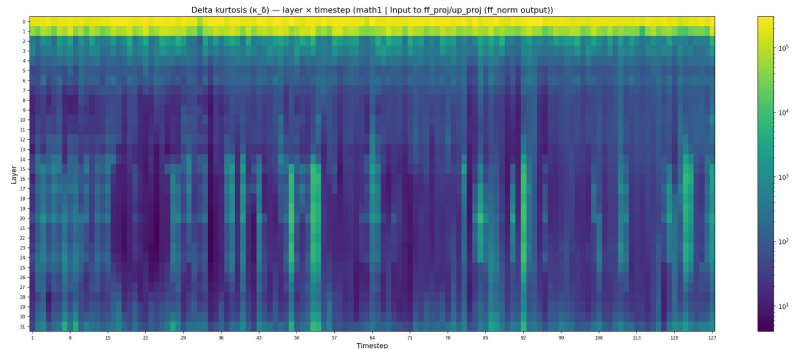
qkv_in



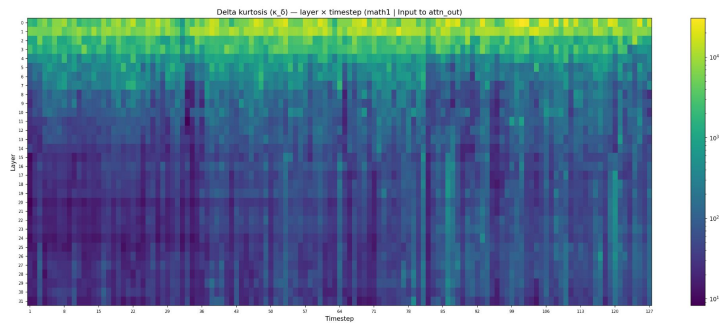
ff_out_in



ff_in

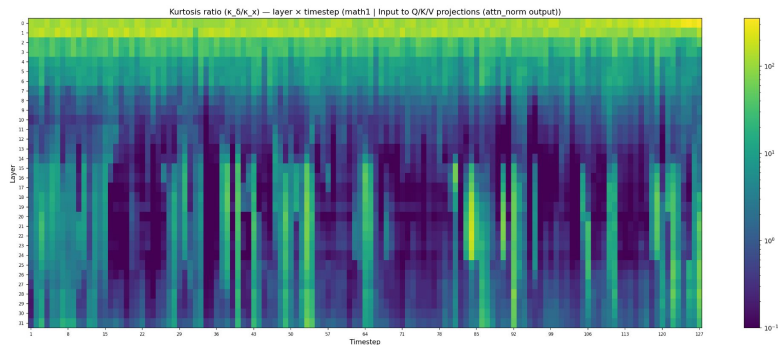


attn_out_in

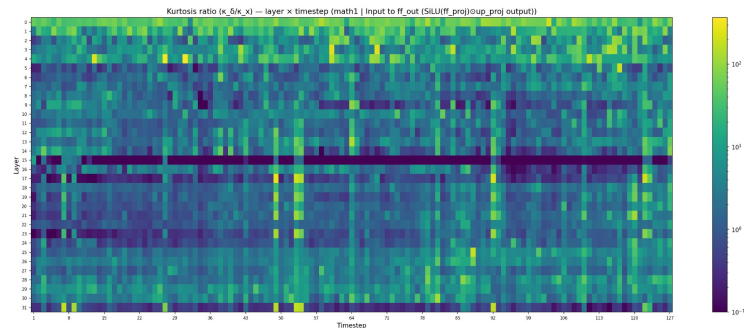


Kurtosis Ratio

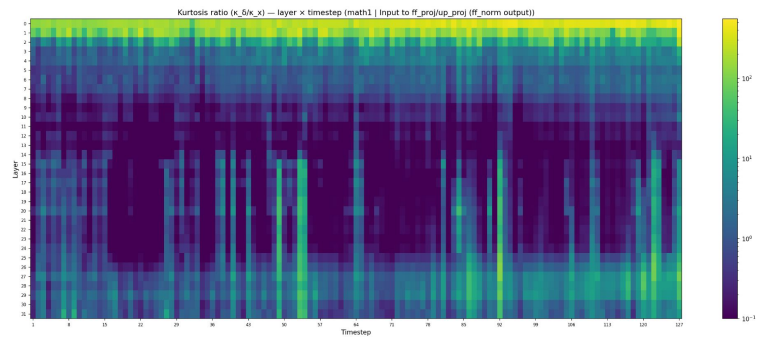
qkv_in



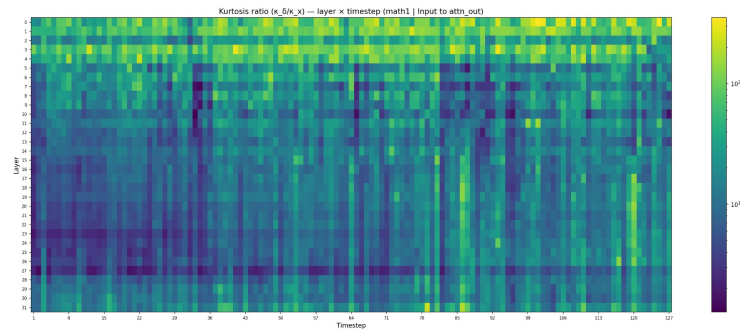
ff_out_in



ff_in



attn_out_in



Delta Quantization

- Activation tensors for a specific submodule and layer does not change significantly from one timestep to the next
- Quantize the difference between activations instead of quantizing the actual activations

$$Y_t^\ell = W^\ell X_t^\ell.$$

$$\Delta X_t^\ell = X_t^\ell - X_{t-1}^\ell \quad X_t^\ell = X_{t-1}^\ell + \Delta X_t^\ell.$$

$$Y_t^\ell = W^\ell (X_{t-1}^\ell + \Delta X_t^\ell) = W^\ell X_{t-1}^\ell + W^\ell \Delta X_t^\ell$$

$$W^\ell X_{t-1}^\ell = Y_{t-1}^\ell$$

$$Y_t^\ell = Y_{t-1}^\ell + W^\ell \Delta X_t^\ell$$

Delta Quantization

- At each timestep we need to store two matrices for each layer:
 - Input activation matrix
 - Output matrix
- The only matrix multiplication computation needed is between the delta and weight, which can be done in INT4 if both matrices are quantized to 4 bits
- The activations at first timestep are kept in full precision
- The cache storing the matrices is updated every timestep and is reseted between different prompts or monte carlo samples

Evaluation

- The likelihood estimation benchmarks do not involve multiple denoising timesteps.
- Generation benchmarks are used to test this approach as they involve multiple timesteps to generate the output:
 - Math: GSM8K, Math
 - Coding: HumanEval, MBPP

Dataset	# fewshots	generation length	diffusion steps	block size
GSM8K	4	256	256	32
Math	0	256	256	64
HumanEval	0	512	512	32
MBPP	3	512	512	32

Results

LLaDA-8B-Base accuracy under different quantization settings. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	24.79	47.58	–	32.32	40.40	36.36	–
	W16A8	AbsMax	68.46	23.64	46.05	↓ 3.21%	33.54	37.60	35.57	↓ 2.17%
		Modified AbsMax	69.52	23.64	46.58	↓ 2.10%	33.54	38.00	35.77	↓ 1.62%
		DeltaQuant	70.43	23.98	47.21	↓ 0.78%	33.54	40.20	36.87	↑ 1.40%
	W4A8	AbsMax + SING	67.93	–	–	–	31.71	38.20	34.96	↓ 3.85%
		DeltaQuant + SING	66.94	–	–	–	33.54	–	–	–

Failure of INT4 Activation

- 8-bit activation

```
[
    " There are 7 x 8 = <<7*8=56>>56 slices of pizza.\nThere are 1 + 3 = <<1+3=4>>4 people who will share the pizza.\nEach of the
m will have 56 / 4 = <<56/4=14>>14 slices of pizza.\n#### 14\n\n"
]
```

- AbsMax 4-bit activation:

[illegible]

- DeltaQuant 4-bit activation:

[illegible]

Plan for Next Week

- Resolve issues with INT4 activation quantization
- Explore if updating the stored copy for activations can improve results
- Find a way to use KL-divergence as a metric for these experiments